

CS 201: Computer Organization

Course Code	CS 201
Course Name	Computer Organization
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG
Distribution	Institute Core
Semester	V
Approval	9th Senate (May 2016)

Course Contents

- **Digital Logic and Data Representation** Introduction to digital logic (logic gates, flip-flops, circuits); Logic expressions and Boolean functions; adder, subtractor, Design of arithmetic and logic unit (ALU)., Representation of numeric data, signed and unsigned arithmetic; Range, precision and errors in floating-point arithmetic; Representation of text, audio and images.
- **Computer Architecture and Organization** History of the digital computer; Introduction to instruction set architecture, micro architecture and system architecture; Instruction sequencing, flow-of-control, subroutine call and return mechanisms; Structure of machine-level programs; Low-level architectural support for high-level languages.
- **Interfacing and I/O strategies** I/O fundamentals: handshaking and buffering; Interrupt mechanisms; vector and prioritized, interrupt acknowledgement; Buses: protocols, arbitration, direct memory access (DMA); Examples of modern buses: e.g. PCIe, USB, Hyper transport.
- **Memory Architecture** Storage systems and their technology, Storage standards (CD-ROM, DVD); Memory hierarchy, latency and throughput; Cache memories-operating principles, replacement policies, multilevel cache, virtual memory system: page table and TLB, cache coherency
- **Functional Organization** Review of register transfer language to describe internal operations in a computer; CISC vs RISC Designs simple implementation schemes datapath design, control unit Micro architectures- hardwired and micro programmed realizations; Instruction pipelining and instruction-level parallelism (ILP); Overview of superscalar architectures; Processor and system performance; Performance their measures and their limitations.

CO Lab

1. The assignment will be designed to assist the theory covered in the class:
2. Some examples are given as follows:
3. Design adder, subtractor in breadboard
4. Designing ALU (Arithmetic and Logical Unit) in breadboard
5. Programming in assembly language and perform some basic operations addition, subtraction, sorting

Text Books

1. C.Hamacher, Z.Vranesic and S.Zaky, **Computer Organization**, 5th Edition, McGraw-Hill, 2002.
2. J.P.Hayes, **Computer Architecture and Organization**, 3rd Edition.
3. D. A. Patterson and J.L.Hennessy, **Computer Organization and Design - The Hardware/Software Interface**
4. William Stallings, **Computer Organization & Architecture designing for performance**, 7th Edition.